

Executive Summary

Comparative Assessment of Nine Agentic Projects

Generated from `report/project_assessment.md`

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Decision Summary

Nine independent project directories were audited against `todo.md`; their full names appear in the ranking and recap tables below. The assignment asked for a PyTorch pedagogical notebook for the three-Dirac stochastic interpolant, comparisons across schedules and optimal transport, a Gaussian-to-Gaussian closed-form analysis with a precise theorem, generated figures, and a polished LaTeX article.

Recommendation. Use `emmanuel/` as the best full research artifact, or `codex-gabriel/` as the best compact PyTorch-first baseline. `openclaw-jeremie/` is the strongest upper-mid repository and is worth mining for PyTorch structure and finite-assignment comparisons, but its Gaussian theory still needs the explicit noncommuting closed form. `hermes-jeremie/` is a clean PyTorch/test/CI submission just below `openclaw-jeremie/`, but its notebooks are unexecuted and its Gaussian theorem remains commuting/time-ordered only. `claudex-gabriel/` and `codex-kimia/` contain useful material, but their Gaussian theory should not be reused without repair. `antigravity-antoine/` is stronger than `antigravity-olivier/` on PyTorch delivery, but remains a thin demonstration rather than a complete solution.

Overall ranking			
Rank	Project	Overall	Executive interpretation
1	<code>emmanuel/</code>	9.1	Best complete research artifact: strongest paper, broadest reproducibility wrapper, passing tests, executed notebooks, and clearest Gaussian endpoint theorem.
2	<code>codex-gabriel/</code>	8.8	Best compact baseline: PyTorch-centered, reproducible, mathematically correct for the central theorem, and supported by lightweight checks.
3	<code>openclaw-jeremie/</code>	8.2	Strongest upper-mid submission: excellent repository, executed notebooks, PyTorch code, source paper, and honest time-ordered Gaussian treatment, but missing explicit general T_t .
4	<code>hermes-jeremie/</code>	7.7	Strong PyTorch repository with passing tests and a cautious Gaussian theorem, but unexecuted notebooks and no explicit general T_t .
5	<code>claudex-gabriel/</code>	7.1	Strong notebooks and paper shell, but the Gaussian theorem is overclaimed and the general noncommuting closed form is missing.
6	<code>codex-kimia/</code>	7.0	Clean compact repository with source and PDFs, but only proves the commuting Gaussian case and has a Torch/OpenMP reproducibility issue.
7	<code>codex-clement/</code>	6.2	Helpful PDF and PyTorch sketch, but no LaTeX source, no dependency file, unexecuted notebooks, and no general T_t .
8	<code>antigravity-antoine/</code>	5.8	Executed PyTorch notebooks and source PDF, but missing dependency/test infrastructure and the closed-form Gaussian T_t .
9	<code>antigravity-olivier/</code>	5.4	Partial visual and mathematical draft, but not PyTorch-based and mathematically unreliable in the noncommuting Gaussian proof.

Score Recap

Detailed scores from the audit

Project	Math	Code	Paper	Goal	Overall	Main limiting factor
antigravity-olivier/	5.5	5.0	5.5	5.5	5.4	NumPy/SciPy instead of PyTorch; flawed noncommuting Gaussian proof; weak reproducibility.
antigravity-antoine/	5.8	6.0	5.5	5.8	5.8	Executed PyTorch notebooks and source PDF, but no dependency file, no tests, weak OT claim, and no closed-form Gaussian T_t .
codex-clement/	7.0	5.5	6.5	6.0	6.2	Missing source/rebuild layer and general noncommuting Gaussian formula.
codex-kimia/	7.0	7.0	7.0	7.0	7.0	Stops at commuting Gaussian theorem; documented figure build failed without unsafe OpenMP workaround.
claudes-gabriel/	6.5	7.5	7.0	7.2	7.1	Good implementation, but global Gaussian iff theorem is stronger than the proof supports.
hermes-jeremie/	7.8	8.0	7.5	7.6	7.7	Clean PyTorch/test/CI repo, but notebooks are unexecuted and Gaussian theory lacks explicit general T_t .
openclaw-jeremie/	8.2	8.6	8.0	8.2	8.2	Excellent PyTorch/repo execution, but no explicit general noncommuting T_t formula.
codex-gabriel/	9.0	9.0	8.0	9.0	8.8	Mostly polish: clarify VP convention, improve float placement, and add stronger environment pinning.
emmanuel/	9.3	8.8	9.0	9.2	9.1	Strongest overall, but main numerical code is NumPy/SciPy-first rather than PyTorch-native.

Requirement Coverage Recap

Requirement-by-requirement coverage									
Requirement	antigravity-olivier/	antigravity-antoine/	codex-clement/	codex-kimia/	claudex-gabriel/	hermes-jeremie/	openclaw-jeremie/	codex-gabriel/	emmanuel/
Three-Dirac notebook	Partial: NumPy/SciPy	Mostly: executed, PyTorch/SciPy	Partial: unexecuted	Mostly: PyTorch, mostly executed	Good: executed, rich	Partial: PyTorch but unexecuted	Good: executed, PyTorch	Good: executed, PyTorch	Good: executed, NumPy/SciPy-first
Closed-form density and velocity	Partial: derived, not PyTorch	Mostly: correct PyTorch formula	Mostly: PyTorch utility	Good: PyTorch	Good: PyTorch	Good: PyTorch module/tests	Good: PyTorch diagnostics	Good: PyTorch checks	Mostly: tested; PyTorch secondary
Three schedules	Partial: VP endpoint issue	Mostly: endpoint-consistent	Mostly: stops before endpoint	Mostly: endpoint clamping	Good: exact endpoints	Good: endpoint tests	Good: endpoint tests	Good: exact endpoints	Good: tested and extended
OT comparison	Partial: linear OT only	Weak: nearest claim not proved	Partial: unexecuted sketch	Mostly: approximate dual	Mostly: symmetric sectors	Partial: greedy OT-like	Good: exact finite balanced	Good: derivation and rays	Good: strongest Laguerre OT
Pedagogical exposition	Partial: thin notebook role	Partial: executed but thin	Partial: concise, unexecuted	Mostly: compact note	Good: strong notebooks	Partial: generated, unexecuted	Good: exposed tensors	Good: strong notebooks	Good: notebooks and audits
LaTeX article and figures	Partial: modest article	Partial: short source/PDF	Partial: PDF only	Mostly: source and PDF	Good: substantial paper	Good: source and PDF	Good: source and clean PDF	Good: strong workflow	Good: best polished paper
Gaussian covariance path Σ_t	Good: correct	Good: correct	Good: correct	Good: correct	Good: correct	Good: correct	Good: correct	Good: correct	Good: correct
General Gaussian T_t	Missing: commuting only	Missing: ODE numerics only	Missing: commuting only	Missing: commuting only	Missing: Euler numerics	Missing: commuting/time-ordered only	Partial: time-ordered only	Good: correct formula	Mostly: endpoint theorem strongest
T_1 vs OT theorem	Partial: flawed proof	Weak: asserted, not proved	Partial: numerical only	Partial: commuting only	Partial: overclaimed	Partial: commuting plus numeric contrast	Mostly: true SPD criterion	Good: correct	Good: clearest explanation
Reproducibility and tests	Weak: no deps/build/tests	Weak: no deps/tests	Weak: no deps/tests/source	Partial: build path, no tests	Partial: no tests	Mostly: lockfile, CI, 5 tests pass	Mostly: strong tests present, not locally run	Good: Makefile and checks	Good: pinned deps, tests, source

Main Risk Recap

Highest-impact audit findings		
Project	Severity	Finding
<code>antigravity-olivier/</code>	High	Does not satisfy the PyTorch requirement and gives an unreliable proof strategy for the noncommuting Gaussian flow map.
<code>antigravity-antoine/</code>	High	Has executed PyTorch notebooks, but no closed-form Gaussian T_t , no proof of the iff OT claim, and weak reproducibility infrastructure.
<code>codex-clement/</code>	High	Has no general closed-form noncommuting T_t , and the submitted repository lacks README, dependency file, LaTeX source, tests, and executed notebooks.
<code>codex-kimia/</code>	High	Correctly proves only the commuting Gaussian case; the requested noncommuting T_t and iff characterization are absent.
<code>claude-gabriel/</code>	High	States a global Gaussian iff theorem but proves only sufficiency plus infinitesimal necessity near commuting covariances.
<code>hermes-jeremie/</code>	High	Clean PyTorch/test repository, but the notebooks are unexecuted and the Gaussian section lacks explicit general T_t and structural iff criterion.
<code>openclaw-jeremie/</code>	High	Strong repo and true time-ordered theorem, but still lacks the explicit closed-form noncommuting T_t requested in <code>todo.md</code> .
<code>codex-gabriel/</code>	Medium	Mostly polish risks: VP naming convention, float placement, and environment pinning.
<code>emmanuel/</code>	Medium	Best overall, but the main numerical implementation is NumPy/SciPy-first despite the PyTorch-centered request.

Mathematical Bottom Line

The central mathematical discriminator is the Gaussian flow-map theorem. The correct noncommuting formula is not obtained by treating the time-varying matrix velocity as if all matrices commute. In the clean formulation, the endpoint map is

$$T_1 = \Sigma_0^{1/2} \left(\Sigma_0^{-1/2} \Sigma_1 \Sigma_0^{-1/2} \right)^{1/2} \Sigma_0^{-1/2},$$

which pushes Σ_0 to Σ_1 . It matches the Gaussian optimal transport map only when this endpoint map is symmetric positive definite, equivalently here when Σ_0 and Σ_1 commute. The full intermediate closed form requires a congruence expression rather than a naive scalar or commuting-matrix simplification. This is exactly where `codex-gabriel/` and `emmanuel/` separate themselves from the middle tier, and where `openclaw-jeremie/` and `hermes-jeremie/` remain one step short despite otherwise strong engineering.

The upper-middle and middle-tier projects are useful for exposition, figures, and numerical intuition, but their Gaussian sections should be repaired before adoption. `antigravity-antoine/` is slightly better than `antigravity-olivier/` because it has executed PyTorch notebooks, but its Gaussian section still stops at ODE numerics and assertion. `antigravity-olivier/` should be treated with even more caution because its proof uses an invalid symmetry argument for a time-dependent matrix ODE and its core notebook path misses the PyTorch requirement.

Recommended Consolidation Plan

1. Select `emmanuel/` as the final paper-first base, or `codex-gabriel/` as the final PyTorch-first base.
2. Import useful PyTorch structure, CI/tests, and finite-assignment or visual ideas from `openclaw-jeremie/`, `hermes-jeremie/`, and `claude-gabriel/`, only after keeping the top-tier Gaussian theorem intact.
3. Reuse `codex-kimia/` or `codex-clement/` material only for compact exposition and figures, not for the Gaussian theorem.
4. Do not import `antigravity-antoine/` or `antigravity-olivier/` formulas without rechecking them against the exact noncommuting covariance calculation and the exact OT comparison.
5. Add one final cross-project regression check: verify endpoint covariance, T_1 equality with Gaussian OT, and schedule independence on noncommuting covariance matrices.